

# Assignment 2 - VHDL Simulation and Synthesis

## Task 1

**Read the paper: Time and delta-delay in VHDL (found under Learning materials on Canvas).**

Done :)

**Explain in your own words what happens in a simulation cycle. You may illustrate your explanation with a flow diagram. Keep your answer short.**

In a simulation cycle, to maintain concurrency, an event queue is created and sorted by the correct order of delta delays.

**Explain what happens to signals that are assigned values with and without explicit delays.**

Explicit vs implicit delay

1. Explicit delay - You specify after how much time the transition triggers
2. Implicit delay - It happens after a delta delay, often  $10^{-15}s = 1 \text{ fs}$

## Task 2

Currently doing it :)

## Task 3

**How many flip-flops are used in the design?**

There are 1370 flip-flops being used in this design.

### **How many LUTs are used in the design?**

There are 284 LUTs being used in this design.

### **What is a LUT?**

**LUT** = Look Up Table

It is a tiny block of memory used to store the truth table of a logic function.

### **What is the total on-chip power?**

The on-chip power is 0.403 Watts.

### **Mention things you can explore in the flowing steps of the design flow (Not sure exactly what is meant here):**

#### **Simulation**

- Add and manage simulation sources (distinguishes between simulation and synthesis files), and configure simulation sets
- Run Behaviour Simulation on the RTL to check functionality
- Explore waveforms to see signals
- Verify design logic

#### **Synthesis**

- Elaborate RLT design (ex: explore RTL netlist, schematic view and so on)
- Browse the IP catalog (customise and generate IP blocks)
- Run RTL level **D**esign **R**ule **C**hecks (DRCs) to catch issues early
- After synthesis, one can explore:
  - Timing constraints and definition

- Different reports (ex. power, components)
- and so on :)

## Implementation

- One can explore
  - Different reports (ex. timing, power ...), now based on actual placed and routed design
  - Actual routed connections on device
  - Generate the final bitstream file for programming the hardware device (FPGA)

RTL → Synthesis → I/O Planning →  
 Placement → Routing → Bitstream

## Hand in answers to the questions

Will do!

## Task 4

The files below are what were created.

`architecture.vhd`

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity latch is
    Port(A,B: in std_ulogic;
          Q, QN: inout std_ulogic := 'Z');
end latch;

architecture behavioral of latch is
begin
    Q <= A nor QN;
```

```
    QN <= B nor Q;  
end behavioral;
```

testbench.vhd

```
library IEEE;  
use IEEE.STD_LOGIC_1164.ALL;  
  
entity test_bench is  
end entity test_bench;  
  
architecture test_latch of test_bench is  
    signal A, B, Q, QN: std_ulogic;  
    begin  
        DUT: entity work.latch(behavioral)  
            port map (A => A, B => B, Q => Q, QN => QN);  
        stimulus: process is  
            begin  
                A <= '1' ; B <= '0' ;  
                wait for 10 ns ;  
                A <= '0' ;  
                wait for 10 ns ;  
                B <= '1' ;  
                wait for 10 ns ;  
                B <= '0' ;  
                wait for 10 ns ;  
                B <= '1' ; A <= '1' ;  
                wait; --suspends the process so it does not loop  
            p :)  
            end process stimulus;  
        end architecture test_latch;
```

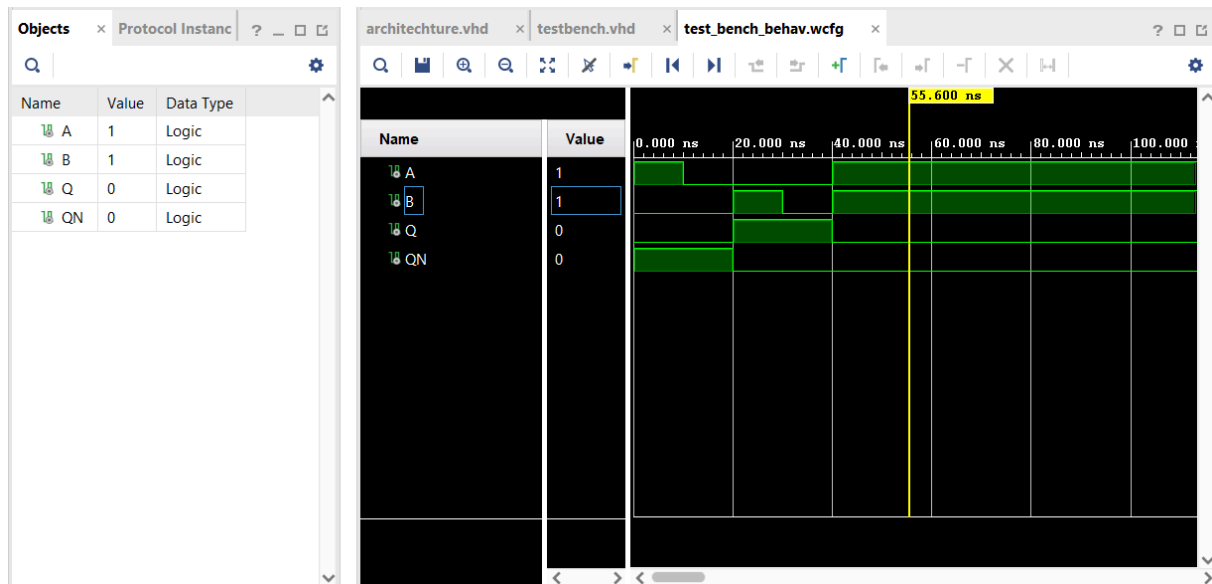
## Simulation of the code

### Does it work as expected?

Yes, seems to work. The output gives us what we expect.

## Does it oscillate?

There is no oscillation. It gets stuck at 1100 (A B Q QN) after 40 ns.



## What would happen if it ran the code sequentially

- When B = 1, after 30 ns, the Q and QN will both be calculated to be 0. This is because Q will use the older value instead of the updated value for QN. This should NOT happen in a latch.
- The rest of the tests would give the same results as in B.

## Seeing delay

We only get errors 😞 We didn't write in VERILOG, see picture, and we could not compile simulation tools required for Questa 😞